

Qualification Pack



Precision Machinist- CNC

QP Code: TNSD/CSC/Q0601

Version: 1.0

NSQF Level: 4

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Qualification Pack

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Qualification Pack

TNSD/CSC/Q0601: Precision Machinist- CNC

Brief Job Description

A Precision Machinist - CNC operates, sets up, and maintains Computer Numerical Control (CNC) machine tools to manufacture precision components according to technical drawings and quality specifications.

Personal Attributes

Upon successful completion, learners will be qualified for employment as a Precision Machinist - CNC or similar roles in engineering, manufacturing, machine tools, and automobile industries. They will contribute to quality production and adapt to evolving CNC technologies.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TNSD/CSC/N0601: Perform Workshop Manufacturing Operations.](#)
2. [TNSD/CSC/N0602: Operate and Implement CNC Programming Techniques](#)
3. [TNSD/CSC/N0603: Simulate CNC Programs using Keller software](#)
4. [TNSD/CSC/N0604: Operate and trouble shoot basic hydraulics and Pneumatics system](#)
5. [TNSD/CSC/N0605: Create 3D Technical Drawings Using AutoCAD](#)
6. [TNSD/CSC/N0606: Create 2D Technical Drawings Using Solid works](#)
7. [TNSD/CSC/N0607: Generate simple CNC Toolpaths using Master cam](#)
8. [TNSD/CSC/N0608: Apply Technical Mathematics for CNC Applications](#)
9. [TNSD/CSC/N0609: Apply basic electrical and sensor in a Manufacturing Environment](#)
10. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC

Qualification Pack

Country	India
NSQF Level	4
Credits	40
Aligned to NCO/ISCO/ISIC Code	7233.0200
Minimum Educational Qualification & Experience	Previous relevant Qualification of NSQF Level (3.5) OR Previous relevant Qualification of NSQF Level (Precision Machinist -Conventional (Level 3.5))
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	1.0
Reference code on NQR	QG-04-CG-04455-2025-V1-TNSDC
NQR Version	1.0

Qualification Pack

TNSD/CSC/N0601: Perform Workshop Manufacturing Operations.

Description

Understand the manufacturing of complex parts by using conventional and CNC lathe, Understand the manufacturing of complex parts by using conventional and CNC Milling Machines. Understand the calculations related to machining

Scope

The scope covers the following :

- Understand the manufacturing of complex parts by using conventional and CNC Machines.
- Understand the different non- conventional machining processes.
- Understand the workshop related calculations.
- Familiarize with machine maintenance process

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the safety aspects and PPE used in CNC Machines
- PC2.** Understand and Demonstrate Special operations – Indexing, Gear cutting, slotting, Sphere milling, Radius milling and Non-conventional Machining process.
- PC3.** Learn different parts and its functions of CNC tool room machines- lathe, Milling.
- PC4.** Understand different modules used in CNC machines
- PC5.** Learn the different types of work holding devices, Tool holding devices of CNC- lathe, Milling.
- PC6.** Learn insert selection as per the raw material.
- PC7.** Demonstrate Part programming and Machining different Exercises on CNC lathe
- PC8.** Demonstrate Part programming and Machining different Exercises on CNC Milling
- PC9.** Understand and Practice the measuring technique by using CMM.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the safety aspects and PPE used in CNC Machines.
- KU2.** Understand and Demonstrate Special operations Indexing, Gear cutting, slotting, Sphere milling, Radius milling and Non-conventional Machining process.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate Part programming and Machining different Exercises on CNC lathe

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	200	-	-
PC1. Understand the safety aspects and PPE used in CNC Machines	-	-	-	-
PC2. Understand and Demonstrate Special operations – Indexing, Gear cutting, slotting, Sphere milling, Radius milling and Non-conventional Machining process.	20	40	-	-
PC3. Learn different parts and its functions of CNC tool room machines- lathe, Milling.	5	20	-	-
PC4. Understand different modules used in CNC machines	5	-	-	-
PC5. Learn the different types of work holding devices, Tool holding devices of CNC- lathe, Milling.	5	-	-	-
PC6. Learn insert selection as per the raw material.	5	-	-	-
PC7. Demonstrate Part programming and Machining different Exercises on CNC lathe	-	60	-	-
PC8. Demonstrate Part programming and Machining different Exercises on CNC Milling	-	60	-	-
PC9. Understand and Practice the measuring technique by using CMM.	10	20	-	-
NOS Total	50	200	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0601
NOS Name	Perform Workshop Manufacturing Operations.
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC, Precision Machinist-CNC
NSQF Level	4
Credits	18
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0602: Operate and Implement CNC Programming Techniques

Description

Able to do the programming and manufacturing of components on CNC lathe. Able to do the programming and manufacturing of components on CNC Milling machine.

Scope

The scope covers the following :

- Understand the concepts and CNC systems
- Familiar with G-codes and M-codes used in programming

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the basic concepts and constructional features of CNC machines
- PC2.** Understand the coordinate system & codes
- PC3.** Learn the concept of offset and different modes of operation
- PC4.** Understand the different types of tool holders and inserts used in CNC machines.
- PC5.** Demonstrate Program and do machining a part on CNC lathe.
- PC6.** Demonstrate Program and do machining a part on CNC Milling.
- PC7.** Fabricate different parts of an assembly/project on CNC machines and assembling

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the basic concepts and constructional features of CNC machines
- KU2.** Understand the different types of tool holders and inserts used in CNC machines.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate Program and do machining a part on CNC lathe.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	90	75	35
PC1. Understand the basic concepts and constructional features of CNC machines	20	-	-	-
PC2. Understand the coordinate system & codes	30	-	-	-
PC3. Learn the concept of offset and different modes of operation	30	-	-	-
PC4. Understand the different types of tool holders and inserts used in CNC machines.	20	-	-	-
PC5. Demonstrate Program and do machining a part on CNC lathe.	-	45	-	5
PC6. Demonstrate Program and do machining a part on CNC Milling.	-	45	-	5
PC7. Fabricate different parts of an assembly/project on CNC machines and assembling	-	-	75	25
NOS Total	100	90	75	35

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0602
NOS Name	Operate and Implement CNC Programming Techniques
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	4
Credits	8
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0603: Simulate CNC Programs using Keller software

Description

Simulation of CNC program by Keller software.

Scope

The scope covers the following :

- Familiar with different tools and functions of KELLER software
- Learn to program on CNC machines.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Practice working environment of CNC machines and identify the parts using keller software.
- PC2.** Do basic programming with right Feed & Speed and understand the Cycle time on CNC lathe using keller software.
- PC3.** Do basic programming with right Feed & Speed and understand the Cycle time on CNC Milling using keller software.
- PC4.** Demonstrate Various ISO cycles and its implementation in CNC Lathe
- PC5.** Demonstrate the Simulation of Converting raw material to finished part on CNC Milling, Lathe using keller software.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Practice working environment of CNC machines and identify the parts using keller software.
- KU2.** Do basic programming with right Feed & Speed and understand the Cycle time on CNC lathe using keller software

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate the Simulation of Converting raw material to finished part on CNC Milling, Lathe using keller software.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	90	-	10
PC1. Practice working environment of CNC machines and identify the parts using keller software.	-	-	-	10
PC2. Do basic programming with right Feed & Speed and understand the Cycle time on CNC lathe using keller software.	-	20	-	-
PC3. Do basic programming with right Feed & Speed and understand the Cycle time on CNC Milling using keller software.	-	20	-	-
PC4. Demonstrate Various ISO cycles and its implementation in CNC Lathe	-	20	-	-
PC5. Demonstrate the Simulation of Converting raw material to finished part on CNC Milling, Lathe using keller software.	-	30	-	-
NOS Total	-	90	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0603
NOS Name	Simulate CNC Programs using Keller software
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0604: Operate and trouble shoot basic hydraulics and Pneumatics system

Description

Understand the basics of hydraulics and pneumatics components. Learn to construct a basic circuits and its functioning.

Scope

The scope covers the following :

- Understand the different types of Pneumatics and Hydraulics components.
- Familiarize with the different symbols used in Pneumatics and Hydraulics circuits.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the different components of Hydraulics and pneumatics system.
- PC2.** List out hydraulic and pneumatic working elements
- PC3.** Design a basic hydraulic and pneumatic circuits
- PC4.** Identify the Hydraulics and Pneumatics components for general and tooling applications
- PC5.** Demonstrate a simple hydraulic circuits & trouble shoot.
- PC6.** Demonstrate a simple pneumatic circuits & trouble shoot.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the different components of Hydraulics and pneumatics system.
- KU2.** Design a basic hydraulic and pneumatic circuits

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate a simple hydraulic circuits & trouble shoot.
- GS2.** Demonstrate a simple pneumatic circuits & trouble shoot.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand the different components of Hydraulics and pneumatics system.	10	-	-	-
PC2. List out hydraulic and pneumatic working elements	15	-	-	-
PC3. Design a basic hydraulic and pneumatic circuits	15	-	-	-
PC4. Identify the Hydraulics and Pneumatics components for general and tooling applications	10	-	-	-
PC5. Demonstrate a simple hydraulic circuits & trouble shoot.	-	20	-	5
PC6. Demonstrate a simple pneumatic circuits & trouble shoot.	-	20	-	5
NOS Total	50	40	-	10

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0604
NOS Name	Operate and trouble shoot basic hydraulics and Pneumatics system
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0605: Create 3D Technical Drawings Using AutoCAD

Description

Able to read and draw CAD drawing assembly and part drawings with GD&T.

Scope

The scope covers the following :

- Learn to draw assembly and detail drawing with AutoCAD.
- Understand to prepare three orthographic views and Isometric view of complicated shapes with dimensioning and sectioning

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Perform Intricate orthographic sections and views with dimensions
- PC2.** Demonstrate assembly drawing with threads and fasteners
- PC3.** Perform drafting assembly views and detailing
- PC4.** Draw views and assemblies incorporating GD&T
- PC5.** Draw machine parts and assembly views

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Perform Intricate orthographic sections and views with dimensions
- KU2.** Demonstrate assembly drawing with threads and fasteners

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Draw views and assemblies incorporating GD&T

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	90	-	10
PC1. Perform Intricate orthographic sections and views with dimensions	-	10	-	-
PC2. Demonstrate assembly drawing with threads and fasteners	-	10	-	-
PC3. Perform drafting assembly views and detailing	-	20	-	10
PC4. Draw views and assemblies incorporating GD&T	-	30	-	-
PC5. Draw machine parts and assembly views	-	20	-	-
NOS Total	-	90	-	10

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0605
NOS Name	Create 3D Technical Drawings Using AutoCAD
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0606: Create 2D Technical Drawings Using Solid works

Description

Able to draw 2D profiles using Solid works software. Able to draw 3D Model using Solid works software.

Scope

The scope covers the following :

- Understand Basics of 2D Drawing, Editing, Modelling (Surface &Solid) by Solid works software.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate the tools for sketching in Solidworks.
- PC2.** Demonstrate the tools for Basic Modelling in Solid works
- PC3.** Perform the method of advanced modelling
- PC4.** Demonstrate the Concept of surface modelling.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Perform the method of advanced modelling
- KU2.** Demonstrate the Concept of surface modelling.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate the tools for Basic Modelling in Solid works

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Demonstrate the tools for sketching in Solidworks.	-	20	-	10
PC2. Demonstrate the tools for Basic Modelling in Solid works	-	20	-	-
PC3. Perform the method of advanced modelling	-	20	-	10
PC4. Demonstrate the Concept of surface modelling.	-	20	-	-
NOS Total	-	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0606
NOS Name	Create 2D Technical Drawings Using Solid works
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0607: Generate simple CNC Toolpaths using Master cam

Description

Simulation of CNC program by Master cam software before machining. Perform machining of a simple irregular profile.

Scope

The scope covers the following :

- Understand the process of creating tool path and NC program
- Understand the process of transferring Program to CNC Milling and Lathe machines.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand screen layout and main menu
- PC2.** Learn to Identify the tools for sketching, modelling
- PC3.** Demonstrate Drawing 2D profiles and Modelling.
- PC4.** Learn to Designing with wire frame
- PC5.** Perform Designing and drawing 3D profile.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn to Designing with wire frame
- KU2.** Learn to Identify the tools for sketching, modelling

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate Drawing 2D profiles and Modelling.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand screen layout and main menu	10	5	-	-
PC2. Learn to Identify the tools for sketching, modelling	10	5	-	10
PC3. Demonstrate Drawing 2D profiles and Modelling.	10	10	-	-
PC4. Learn to Designing with wire frame	10	10	-	-
PC5. Perform Designing and drawing 3D profile.	10	10	-	-
NOS Total	50	40	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0607
NOS Name	Generate simple CNC Toolpaths using Master cam
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0608: Apply Technical Mathematics for CNC Applications

Description

Understand the concept of mathematics used for doing calculations related to engineering application and general purpose.

Scope

The scope covers the following :

- Understand the applications of Trigonometrical functions in engineering field.
- Understand the basic concept & application of calculus.
- Understand the concept of Analytical Geometry.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn the Concepts of fundamental Analytical geometry
- PC2.** Find the slope of line , parallel and perpendicular lines
- PC3.** Find the distance between the vertices Transversal of line
- PC4.** Calculate the area of triangles and quadrilaterals , mid points, Equation of lines and related problems.
- PC5.** Calculate the Centroid , in center, Ortho center of a triangles and related problems.
- PC6.** Understand the basic concept & application of calculus
- PC7.** Solve the linear and non-linear equations , Ordinary differential & partial differential equations.
- PC8.** Find the order and degree of the differential equations, solution of ODE& PDE.
- PC9.** Learn the Differential rule , Chain rule, Bernoulli rule and related problems
- PC10.** Learn Basic concept of principle of Trigonometry and its applications
- PC11.** Solve the problems using trigonometry application
- PC12.** Find the Solution of acute , obtuse angled triangles and related problems
- PC13.** Find the values of angles by using trigonometric table values, Angle value for the special angles, Solution of circumscribed triangles by sine rule and practice problems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn the Differential rule , Chain rule, Bernoulli rule and related problems
- KU2.** Understand the basic concept & application of calculus

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Solve the linear and non-linear equations , Ordinary differential & partial differential equations.
- GS2.** Find the values of angles by using trigonometric table values, Angle value for the special angles, Solution of circumscribed triangles by sine rule and practice problems.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Learn the Concepts of fundamental Analytical geometry	5	-	-	-
PC2. Find the slope of line , parallel and perpendicular lines	5	-	-	-
PC3. Find the distance between the vertices Transversal of line	5	-	-	-
PC4. Calculate the area of triangles and quadrilaterals , mid points, Equation of lines and related problems.	10	-	-	-
PC5. Calculate the Centroid , in center, Ortho center of a triangles and related problems.	10	-	-	-
PC6. Understand the basic concept & application of calculus	5	-	-	-
PC7. Solve the linear and non-linear equations , Ordinary differential & partial differential equations.	10	-	-	-
PC8. Find the order and degree of the differential equations, solution of ODE& PDE.	10	-	-	-
PC9. Learn the Differential rule , Chain rule, Bernoulli rule and related problems	10	-	-	-
PC10. Learn Basic concept of principle of Trigonometry and its applications	5	-	-	-
PC11. Solve the problems using trigonometry application	10	-	-	-
PC12. Find the Solution of acute , obtuse angled triangles and related problems	10	-	-	-
PC13. Find the values of angles by using trigonometric table values, Angle value for the special angles, Solution of circumscribed triangles by sine rule and practice problems.	5	-	-	-

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Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	100	-	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0608
NOS Name	Apply Technical Mathematics for CNC Applications
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0609: Apply basic electrical and sensor in a Manufacturing Environment

Description

Understanding the applications of basic electrical circuits Understanding the applications of sensors.

Scope

The scope covers the following :

- Understand the construction of electric circuits.
- Understand the concepts of sensors

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Basic of Electricity - Voltage -Current

PC2. Learn Electromagnetic Induction Principle-Working of electrical components like Solenoids, Motors, Generator, Switches, Fuse and Circuit Breakers

PC3. Learn the Fundamentals of Sensor and its types

PC4. Understand the Application of sensors

PC5. Understand the basic Electricals and Electronics components for Machine Tool applications

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Basic of Electricity ,Voltage ,Current

KU2. Understand the Application of sensors

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Learn Electromagnetic Induction Principle-Working of electrical components like Solenoids, Motors, Generator, Switches, Fuse and Circuit Breakers

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	-	-	-
PC1. Understand the Basic of Electricity - Voltage -Current	10	-	-	-
PC2. Learn Electromagnetic Induction Principle-Working of electrical components like Solenoids, Motors, Generator, Switches, Fuse and Circuit Breakers	10	-	-	-
PC3. Learn the Fundamentals of Sensor and its types	10	-	-	-
PC4. Understand the Application of sensors	10	-	-	-
PC5. Understand the basic Electricals and Electronics components for Machine Tool applications	10	-	-	-
NOS Total	50	-	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0609
NOS Name	Apply basic electrical and sensor in a Manufacturing Environment
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping TNSDC
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- TNSDC monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

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- Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored

On the Job:

1. Each module will be assessed separately.

2. The candidate must score 60% in each module to successfully complete the OJT.

3. Tools of Assessment that will be used for assessing whether the candidate is having desired skills and etiquette of dealing with customers, understanding needs & requirements, assessing the customer and perform Soft Skills effectively:

- Videos of Trainees during OJT

4. Assessment of each Module will ensure that the candidate is able to:

- Effective engagement with the customers
- Understand the working of various tools and equipment

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TNSD/CSC/N0601.Perform Workshop Manufacturing Operations.	50	200	-	-	250	15
TNSD/CSC/N0602.Operate and Implement CNC Programming Techniques	100	90	75	35	300	20
TNSD/CSC/N0603.Simulate CNC Programs using Keller software	-	90	-	10	100	10
TNSD/CSC/N0604.Operate and trouble shoot basic hydraulics and Pneumatics system	50	40	-	10	100	10
TNSD/CSC/N0605.Create 3D Technical Drawings Using AutoCAD	-	90	-	10	100	10
TNSD/CSC/N0606.Create 2D Technical Drawings Using Solid works	-	80	-	20	100	10
TNSD/CSC/N0607.Generate simple CNC Toolpaths using Master cam	50	40	-	10	100	10
TNSD/CSC/N0608.Apply Technical Mathematics for CNC Applications	100	-	-	-	100	5
TNSD/CSC/N0609.Apply basic electrical and sensor in a Manufacturing Environment	50	-	-	-	50	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	420	660	75	95	1250	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.